TSM 098 Arithmetic Test

Practice 1 | Version 202610122020.1

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Evaluate at x = -2, y = 3. Show the substituted expression. $3x - 4y - 2$	1. Value 20 pts
2. Evaluate each side separately at x = 2, -2, 0. Record the left value, right value, and whether they are equal. State what the tested values establish; do not solve the equation. $x(x+2) = x^2 + 2x$	2. x = 2: left / right / equal? 4 pts x = -2: left / right / equal? 4 pts x = 0: left / right / equal? 4 pts Conclusion 8 pts
3. Evaluate each expression. $(a)\quad 9 - (-5)$ $(b)\quad (-6)(-4)$ $(c)\quad \frac{7}{0}$ $(d)\quad -8 - 5$	3. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts
4. Evaluate the same expression at (a) x = 10, (b) x = -10, and (c) x = 0. Show substitution. $x^2 - 3x + 2$	4. (a) 7 pts (b) 7 pts (c) 6 pts
5. Reduce each fraction to lowest terms. $(a)\quad \frac{14}{21}$ $(b)\quad \frac{42}{66}$	5. (a) 10 pts (b) 10 pts

Instructor answer key: TSM 098 Arithmetic Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Value

-20

20 pts

2. x = 2: left / right / equal?

Left 8; right 8; yes

4 pts

x = -2: left / right / equal?

Left 0; right 0; yes

4 pts

x = 0: left / right / equal?

Left 0; right 0; yes

4 pts

Conclusion

All tested values work; this alone does not prove an identity.

8 pts

3. (a)

14

5 pts

(b)

24

5 pts

(c)

Undefined

5 pts

(d)

-13

5 pts

4. (a)

72

7 pts

(b)

132

7 pts

(c)

2

6 pts

5. (a)

$\frac{2}{3}$

10 pts

(b)

$\frac{7}{11}$

10 pts